

BlockFedEDAuth-R

Drift-Triggered, Blockchain-Anchored Rollback for Federated Anomaly Detection in Semiconductor Defect Inspection

A Final-Year Project Report: System Design, Related Work, Open Decisions, and Research Directions

Prepared as a working project specification and literature-grounded design document, compiled from an extended design discussion.

How to read this document. This report consolidates a design discussion into a structured specification. Every major design choice is presented as an explicit decision point with options and trade-offs rather than a single prescribed answer, because several choices are still open. Sections 6 and 9 are the most important for planning: Section 6 lays out the specific research directions available at each layer of the system, and Section 9 is an unvarnished assessment of scope, novelty, and feasibility for a single-student final-year timeline.

1. Executive Summary

The starting point for this project is a recent paper, *FedEDAuth: Federated Embedding Distribution Authentication for Counterfeit IC Detection* (Lodge et al., 2026), which filters malicious clients out of a federated learning (FL) pipeline for counterfeit integrated-circuit detection, before their poisoned updates can be aggregated into the global model. The original evaluation used 64× 64 pixel package-level photographs and a supervised authentic-vs-counterfeit classifier.

Over the course of this design discussion, the project direction shifted in three material ways, each of which is carried through this report:

- **Vision task:** moved away from package/label classification (assessed as a comparatively shallow vision problem for a research-grade project) toward **semiconductor surface defect anomaly detection** on higher-resolution microscopy imagery — a harder, more defensible, and more data-rich problem.
- **Central research contribution:** rather than re-implementing FedEDAuth's authentication math as the novelty, the project's core evaluated claim is now a **cross-round, drift-triggered rollback mechanism:** detecting when the global model has quietly degraded over several rounds (not just within one round) and reverting to a prior good checkpoint, anchored via blockchain-recorded hashes rather than on-chain weights.
- **Blockchain's role** was re-scoped away from being a decorative audit log toward a small number of places where it is argued to add real value: decentralizing trust away from a single authentication server, staking/slashing as an economic deterrent, dispute resolution for false positives, and a device-provenance passport layer downstream of the FL system.

This report is organized so that each layer of the system — dataset/vision, FL aggregation, authentication, rollback/versioning, culprit attribution, and blockchain — is discussed with its relevant prior work, the decision(s) still open, and the specific research question that could be tackled there. Section 9 gives a candid assessment of which of these are realistic to pursue in depth within a single final-year project, and recommends a scoped minimum viable system.

2. Problem Statement and Motivation

Counterfeit and defective components entering the semiconductor supply chain are a persistent, costly, and safety-relevant problem. Federated learning is attractive for supply-chain-wide defect/counterfeit detection because it lets mutually distrustful organizations (fabs, distributors, assembly houses) jointly improve a shared detection model without pooling proprietary imagery. However, FL's open participation model creates an attack surface: a single compromised or careless participant can inject poisoned updates that degrade the shared model, and this can happen slowly and subtly enough to evade single-round detection.

Two compounding weaknesses motivate this project specifically:

- Existing per-round defenses (FedAvg, trimmed mean, Krum, and FedEDAuth's own embedding-based authentication) evaluate each client's contribution largely in isolation, round by round. None of them are designed to catch a slow, low-intensity drift that stays under each round's detection threshold but accumulates into real damage — a failure mode the FedEDAuth paper itself observed at its lowest tested poisoning intensity (50%), where one poisoned client began to overlap with the clean cluster.

- Once a poisoned update is aggregated, there is no way to 'undo' it from the model's weights directly. The only recourse is recovery: rolling back to, or reconstructing, a known-good prior state. This is a distinct research problem from detection, and is comparatively less mature in the literature when combined with a decentralized, multi-organization trust setting.

The project therefore targets the intersection of three research threads that individually exist in the literature but are not, to the depth searched for this report, combined the way this project proposes: (i) embedding-distribution-based client authentication, (ii) checkpoint-based poisoning recovery, and (iii) blockchain-anchored accountability for federated learning.

3. Related Work

3.1 Federated learning for hardware/IC security

FedEDAuth (Lodge et al., 2026) and its predecessor byzantine-poisoning study (Lodge, Tambe, Aklekar & Saqib, IEEE PAINE 2025) establish that federated learning can reach over 94% accuracy on counterfeit IC classification while remaining vulnerable to a stealthy scratch-trigger poisoning attack that evades FedAvg, trimmed-mean, and Krum aggregation. FedEDAuth's fix is a separate authentication server that scores each client's embedding distribution (from a fixed, shared Vision Transformer) against a golden reference using outlier fraction, class mean shift, and a micro-cluster score, filtering suspicious clients before aggregation. The paper explicitly names its own centralization of this authentication server as an open weakness.

3.2 Sequential vs. parallel federated learning

FedAvg (McMahan et al., 2017) is the standard parallel aggregation scheme: clients train independently and simultaneously on a shared model snapshot, and the server averages their resulting weights. An alternative, sequential family exists — Cyclic Weight Transfer (Chang et al., 2018) and related serial FL methods — where a single model is passed from client to client, each training on top of the last. The literature is consistent that serial/sequential methods are more compute-efficient per round but are meaningfully more prone to catastrophic forgetting and are sensitive to client visit order, especially under non-IID data (Cyclical Weight Consolidation, 2024; Sharp Bounds for Sequential Federated Learning on Heterogeneous Data, 2025). This project currently targets **FedAvg** as the aggregation scheme (see Section 5.2), with sequential/CWT-style training noted as a possible future direction rather than the current design.

3.3 Recovering / rolling back federated models after poisoning

This is the most directly relevant prior-art cluster to the project's proposed central contribution:

- **FedRecover** (Cao, Jia, Zhang & Gong, IEEE S&P; 2023) was among the first to formalize recovering an accurate global model after malicious clients are identified, using the server's stored historical update information rather than retraining from scratch.
- **Crab** ("Towards Efficient and Certified Recovery from Poisoning Attacks in Federated Learning", 2024) improves on this with *selective* historical information and *adaptive* rollback to a historical model that was not significantly affected by malicious clients (rather than the original model), with a theoretical bound on how far the recovered model differs from a full retrain.

- **PDLRecover** (2025) extends recovery to a decentralized, privacy-preserving setting using machine unlearning, and **FAUN** (“Adversarial Update-Based Federated Unlearning for Poisoned Model Recovery”, 2026) proposes a lightweight adversarial-optimization approach that needs only a short window of malicious updates rather than full history.
- **DP-RTFL** (Talluri, 2025) introduces a “Temporal Checkpoint Manifold” — a distributed, chronologically ordered log of global model states enabling precise rollback — paired with lightweight hash-based integrity commitments rather than full zero-knowledge proofs.
- **Leash-FL** (“Lightweight ECC-Based Self-Healing Federated Learning Framework for Secure IIoT Networks”, Sensors, 2025) is the closest published system to this project's proposal: it combines lightweight client authentication with **blockchain-backed checkpoint rollback**, reporting over 85% accuracy retained with 50% malicious clients and recovery up to 3× faster than anomaly-screening-only baselines.

Implication: blockchain-anchored FL checkpoint rollback is not a novel idea in isolation — Leash-FL already published it, with strong results. Any submission built around this must explicitly differentiate itself (Section 6.2 proposes where that differentiation can come from) rather than presenting rollback itself as the contribution.

3.4 Blockchain for FL accountability

BlockFLA (Desai, Ozdayi & Kantarcioglu, ACM CODASPY 2021) is the closest prior art to the project's blockchain-authentication idea: a hybrid blockchain architecture (combining public/Ethereum-style and private/Hyperledger-style components) in which smart contracts automatically detect and punish attackers via monetary penalties, evaluated with both FedAvg and SignSGD aggregation. Reviewers/participants can report a suspected malicious model, triggering a penalty smart contract that excludes it from aggregation and holds its submitter accountable. A 2022 survey (Lo et al.) describes a related trustworthy FL framework focused on accountability and equitable reward distribution. **Implication:** the “blockchain punishes bad actors” component of this project's design is also not new; BlockFLA must be cited and differentiated from directly.

3.5 Datasets

Two dataset families were evaluated for the vision task and are compared in Section 5.1: the MIIC dataset (Huang et al., ICIP 2021, NTU) — 512×512 grayscale SEM images of the wire-bond/metal layer, only 116 anomalous of 25,276 total — and the newly released IDA/ICME 2026 Grand Challenge dataset (arXiv:2607.04675, 2026), which is substantially larger and richer.

4. Proposed System Overview

The system, provisionally named **BlockFedEDAuth-R** (the ‘-R’ for its rollback focus), has six layers. Each is expanded with its own decision points in Section 5.

#	Layer	Role	Status
1	Vision / anomaly model	Detects surface defects from microscopy images on each client's local data.	Open — dataset TBD (5.1)
2	FL aggregation	Combines client updates into a shared global model.	Decided: FedAvg (5.2)
3	Client authentication	Screens client embeddings for statistical poisoning signs before their update is trusted.	Open — needs one-class adaptation (5.3)
4	Rollback & versioning	Detects multi-round drift in the global model and reverts to a prior good checkpoint.	Core contribution (5.4)

#	Layer	Role	Status
5	Culprit attribution	Identifies which client(s) most likely caused a detected drift.	Supporting, scoped down (5.5)
6	Blockchain layer	Decentralizes trust, anchors checkpoint hashes, handles staking/slashing, disputes, and provenance.	Open — platform TBD (5.6)

Data flow, in brief: each client trains locally on its private defect imagery (Layer 1) and participates in FedAvg rounds (Layer 2). Before each round's update is trusted, its embeddings are screened (Layer 3). Independently of per-round screening, the global model's own behavior on a held-out reference set is tracked across a sliding window of rounds (Layer 4); when it drifts beyond a threshold, the system rolls back to the last checkpoint before the drift began, and a lightweight windowed check (Layer 5) narrows down which recent client(s) most likely caused it. All checkpoint hashes, authentication decisions, rollback events, and (if pursued) staking/slashing actions are recorded through the blockchain layer (Layer 6), which never stores model weights directly.

5. Component-by-Component Design and Decision Points

5.1 Vision task and dataset

Decision made: the vision task is semiconductor surface/microscopy defect detection, not package-label counterfeit classification. This is a harder, one-class-learning problem (defects are rare relative to normal samples) rather than a roughly-balanced binary classification problem, which changes the client-side model family from a classifier (e.g. ResNet-18) toward reconstruction-based or embedding-distance-based anomaly detectors (autoencoders, normalizing flows, or distance-to-reference scoring similar to what FedEDAuth already uses for authentication, but now applied to the task itself).

Dataset options

Dataset	Modality / size	Anomalies	Fit for this project
MIIC (Huang et al., ICIP 2021, NTU)	512×512 grayscale SEM, wire-bond/metal layer, 25,276 images total	116 anomalous — extremely imbalanced	Confirmed public and directly matches the anomaly-detection framing, but very few positive examples limits how much can be learned or evaluated per simulated client; splitting 116 anomalies across several non-IID clients leaves very little signal per client.
IDA / ICME 2026 Grand Challenge (arXiv:2607.04675, 2026)	High-resolution wafer-surface microscopy, 7 defect categories (scratch, dent, particle, damage, stain, bubble, chipping), pixel-level instance masks	Track 1: 1,173 anomalous in training (2,543 instances) + 1,834 anomalous in test (1,921 instances) — roughly 10× more anomalous images than MIIC	Much richer: multiple defect categories (useful for simulating different clients seeing different defect distributions — natural non-IID), severity grading as a bonus task, and deliberately multi-scenario construction. Public download access for the full dataset was not confirmed at time of writing and should be verified directly with the challenge organizers before committing to it.

Dataset	Modality / size	Anomalies	Fit for this project
MVTec-AD / ZJU-Leaper (general industrial anomaly benchmarks)	Varies; widely used in anomaly-detection literature	Small per category, similar imbalance profile to MIIC	Useful as an additional cross-domain baseline or for prototyping the anomaly-detection pipeline before committing to a semiconductor-specific dataset, but not semiconductor-specific.
WM-811K / Mixtype-WM38 (wafer bin/defect maps)	811K+ categorical wafer bin maps (pattern shapes across a wafer), not photographic	Large and well-balanced across pattern classes	Different modality entirely (structured pattern maps, not microscopy photos) — useful only if the project pivots to wafer-map pattern classification rather than surface-image defect detection.

Open decision: confirm public access to the IDA/ICME 2026 dataset; if unavailable, fall back to MIIC with an explicit, stated limitation about anomaly count, or combine MIIC with synthetic augmentation (similar to how the FedEDAuth paper synthetically produced its counterfeit indicators) to increase usable positive samples per simulated client.

5.2 Federated learning aggregation

Decision made: FedAvg (parallel aggregation), not Cyclic Weight Transfer, as the core scheme, pending any concrete evidence found during the project that CWT outperforms FedAvg for this specific task and data distribution. This avoids CWT's documented catastrophic-forgetting and order-sensitivity risk under non-IID data, at the cost of losing CWT's “one clean checkpoint per client” property — meaning checkpointing (Section 5.4) will be per-round rather than per-client, and culprit attribution (Section 5.5) cannot rely on the model literally being that client's exact output.

Because real fab data is close to non-IID by construction (different clients naturally seeing different defect category distributions), plain FedAvg's known weakness under heterogeneous data is directly relevant here. Established mitigations worth evaluating as part of the FedAvg baseline, not as the paper's core contribution, include FedProx (proximal-term regularization to limit client drift) and server-side momentum variants (FedAvgM). These are candidate research-direction items in Section 6.1.

5.3 Client authentication (adapted FedEDAuth)

FedEDAuth's suspicion-score math (outlier fraction via Mahalanobis distance, class mean shift, micro-cluster score) was built around a labeled, multi-class golden reference. In a one-class anomaly-detection setting, there typically is no clean, separate “counterfeit” class to compute a reference distribution against — only a “normal” class and rare, diverse anomalies. Adapting this is itself a concrete piece of work: candidate approaches include computing the reference distribution only over the normal class and treating anomaly-embedding statistics separately, or reframing outlier/shift/cluster metrics relative to the client's own reconstruction-error distribution rather than a classification embedding.

Open decision: whether this adaptation is developed as its own evaluated sub-contribution or implemented at “working, not deeply benchmarked” depth, per the scope discussion in Section 9.

5.4 Rollback and versioning — core contribution

What triggers a rollback

Rather than only asking “is this client suspicious this round” (FedEDAuth's scope), the global model's own performance/embedding behavior on the golden reference set is tracked across a sliding window of rounds. A sustained drift in outlier fraction, mean shift, or reconstruction error over that window — even if no single round crossed the per-round threshold — triggers a rollback to the last checkpoint recorded before the drift began. This is designed specifically to catch the low-intensity poisoning case the FedEDAuth paper itself found could evade per-round detection at high poisoning density.

How checkpoints are stored efficiently

- **Never store full weights on-chain.** On-chain storage cost scales with bytes written and is priced punitively; a full model checkpoint (tens of MB) is not merely expensive but categorically infeasible on-chain on any general-purpose chain.
- **Store only a hash + round number + metrics per round on-chain** (a fixed, small cost regardless of model size). This is the pattern used uniformly across the rollback/recovery literature reviewed in Section 3.3 (DP-RTFL, Leash-FL).
- **Store the actual weights off-chain** (IPFS, or another content-addressed / conventional store). Content-addressing gives tamper-evidence for free: the retrieval hash is recomputed and compared against the on-chain record, so integrity does not require also storing the payload on-chain.
- **Reduce storage cost further with delta/differential checkpointing** — store only what changed relative to the previous checkpoint rather than a full snapshot every round, materializing full snapshots only periodically as recovery anchors.
- **Structure each checkpoint as a Merkle tree over layers/parameter groups** rather than one flat hash, so that verification, delta computation, and (potentially) partial/layer-level rollback can operate on just the affected branch rather than the whole model.
- **Prune on a sliding window.** Once a checkpoint is fully superseded and no longer a plausible rollback target, discard its full off-chain payload and keep only its on-chain hash as a historical record.

Evaluation questions for this component

- Does drift-triggered (multi-round) rollback recover accuracy after a low-intensity attack that per-round-only filtering misses, and how much faster/cheaper is this than retraining from scratch or than full-history recovery methods (FedRecover-style)?
- How does detection latency (rounds elapsed before rollback triggers) trade off against the drift threshold's false-positive rate on benign, naturally-noisy non-IID clients?
- What is the storage/communication overhead of the hash-chain + delta-checkpoint scheme relative to naive full-checkpoint-every-round and relative to Crab's selective-history approach?

5.5 Culprit attribution — scoped down, supporting role

Full Shapley-value-based contribution attribution (a well-studied but expensive and, per a 2025 stability study found during this project's literature search, somewhat volatile technique) is not recommended as a project focus — it is its own deep research area and running it continuously across all training history is unnecessary for this project's goals. Instead, attribution is scoped to only the small window of rounds flagged by a rollback trigger (Section 5.4): a cheap, distance-based similarity score (how closely a client's submitted update direction matches the direction the global model drifted in) is computed only for clients active in that window. This directly reuses the rollback trigger to make an otherwise-expensive problem cheap, and is one of the more genuinely distinguishing design choices available to this project (see Section 6.3).

5.6 Blockchain layer

Where blockchain is argued to add real value (and where it does not)

Based on the design discussion, blockchain does **not** meaningfully strengthen the FL math itself — a rejected or accepted client contributes (or doesn't) the same amount to the model whether the decision is logged in a normal database or an immutable ledger. Its genuine value in this system is scoped to four specific jobs:

Mechanism	What it actually does	Would a normal database suffice instead?
Decentralized authentication trust	Removes the need for any single participating company to host (and be implicitly trusted to operate honestly) the authentication server, via multiple independent validators agreeing before a client is accepted or rejected.	No — this is the one place a single trusted-server database cannot substitute, because the whole point is removing that single trusted party among mutually distrustful competitors.
Staking / slashing	Economic deterrent: participants post collateral that is forfeited on confirmed malicious behavior, shifting the attacker's cost-benefit calculation before an attack, not after.	Partially — the economic mechanism itself needs a settlement layer with real value at stake; a plain database cannot enforce forfeiture without a trusted operator.
Dispute resolution / non-repudiation	Signed, multi-party-witnessed record lets competing organizations resolve blame (e.g., whose update caused a rollback) without relying on one side's internal logs.	No — the value here is specifically that no single party's database is treated as ground truth.
Device / component provenance passport	Tracks a verified chip's status across foundry → distributor → integrator → end customer, a genuine multi-party, no-single-authority provenance problem.	No — this is blockchain's best-established real-world use case and the strongest natural fit in this whole architecture.
Plain audit logging of every round	Recording routine, non-disputed events with no economic or multi-party trust stakes.	Yes — this is the “namesake blockchain” trap; a signed, append-only log in a normal database achieves the same practical outcome with far less overhead.

Handling false positives on an append-only ledger

Since entries cannot be deleted, mistaken rejections/slashes are handled with a provisional-flag-plus-challenge-window pattern (analogous to optimistic-rollup dispute periods): a failed authentication produces a provisional flag, not an instant final slash; a challenge window allows the flagged client or other validators to dispute it; only an unchallenged or upheld flag finalizes. The original flag is never erased — a later “overturned” transaction is appended alongside it, so the ledger's current state reflects the correction without rewriting history.

Platform choice: public/Ethereum-style vs. permissioned/Hyperledger-style

Dimension	Public / Ethereum-style (testnet)	Permissioned / Hyperledger-style
Who can join / validate	Open (or open-ish on testnet); anyone can run a node.	Closed consortium — only approved organizations (fabs, distributors) run nodes; matches this project's actual multi-company scenario more directly.
Cost model	Real (or testnet-simulated) gas fees per transaction; storage and computation are priced to discourage bloat — relevant context for justifying the hash-only, off-chain-weights design.	No public gas market; costs are the consortium's own infrastructure, so per-transaction economics are not a natural fit for demonstrating “why not store weights on-chain” unless deliberately modeled.

Dimension	Public / Ethereum-style (testnet)	Permissioned / Hyperledger-style
Staking / slashing token economics	Native to the ecosystem — ERC-20 style stake tokens and slashing are well-trodden patterns with existing tooling and prior academic examples (BlockFLA itself used a hybrid public/private architecture).	Requires bespoke design — permissioned chains do not have a default token economy, so staking would need custom implementation (e.g., a consortium-issued token or off-chain escrow with on-chain attestation).
Throughput / latency	Public testnets are slower and less predictable.	Generally faster and more predictable for a small, known validator set — Leash-FL specifically reports its permissioned-style layer outperforming Ethereum-based systems on latency/throughput.
Fit for a student project timeline	Large existing tooling ecosystem (Solidity, Hardhat/Truffle, public testnets like Sepolia) makes a working prototype faster to stand up.	More realistic to the actual multi-company scenario, but more implementation overhead (Fabric chaincode, network setup) for a solo/small-team timeline.

Recommendation for a first working prototype: build on a public Ethereum-style testnet for speed of development and tooling maturity, explicitly note in the report that a permissioned architecture (as BlockFLA itself uses in hybrid form, and as Leash-FL argues performs better) would be the production-appropriate choice for real rival semiconductor companies, and treat that migration as future work rather than something that needs to be built now.

6. Research Directions, Mapped to Each Open Question

This section answers directly: at which specific points can this project bring a genuine improvement over existing work, organized by the four areas raised during the design discussion.

6.1 Improvements in analyzing / adapting FedAvg

- Quantify how FedAvg's known non-IID sensitivity specifically interacts with the drift-detection signal used for rollback — does heterogeneity across clients widen the 'normal drift' band enough to hide real poisoning-induced drift, or make benign clients look falsely suspicious? This is a concrete, testable question the reviewed literature does not appear to answer directly.
- Evaluate whether a lightweight non-IID mitigation (FedProx-style proximal regularization) changes the drift-detector's false-positive rate, as an ablation rather than a headline contribution.
- Compare plain FedAvg against FedAvg with the adapted one-class FedEDAuth authentication layer (Section 5.3) as a baseline table, to establish how much the authentication layer alone buys before rollback is even introduced.

6.2 Improvements in how rollback occurs (the project's proposed core claim)

- Multi-round drift detection as the rollback trigger, versus the existing literature's approach of triggering recovery only after a client has already been identified as malicious by a separate detector (FedRecover, Crab) — this project's angle is to *use the authentication layer's own statistics, tracked over time*, as the trigger itself, rather than treating detection and recovery as fully separate stages.
- Direct empirical comparison against Crab's selective-history approach and Leash-FL's similarity-governed screening: does window-based drift triggering recover faster, cheaper, or more accurately under the specific low-intensity poisoning regime the base FedEDAuth paper flagged as its own weak spot?
- Explicit differentiation from Leash-FL required: Leash-FL already does blockchain-backed checkpoint rollback with ECC-based lightweight authentication. This project's differentiator, if pursued, is the *drift-over-a-window* trigger condition tied specifically to the FedEDAuth-style suspicion metrics, applied to a one-class anomaly-detection task rather than Leash-FL's intrusion-detection/IIoT setting.

6.3 Improvements in identifying the culprit client

- Rather than general-purpose Shapley-value attribution (expensive, and shown to be volatile in recent work), scope attribution to only the rounds inside a triggered rollback window, using a cheap similarity-to-drift-direction score. Evaluating whether this windowed, cheap approach identifies the correct culprit as reliably as full Shapley-based methods, at a fraction of the compute, is a concrete, testable research question that appears open in the literature reviewed here.
- If time allows, compare against a lightweight Shapley approximation (e.g. Monte Carlo sampling) computed only within the same restricted window, to isolate whether the benefit comes from the windowing itself or from abandoning Shapley altogether.

6.4 Where blockchain logging is genuinely significant (not namesake)

- **Decentralized authentication consensus:** replacing FedEDAuth's single trusted authentication server with a small multi-validator scheme (e.g., a threshold of independent validators must agree on a client's suspicion score before it is accepted or rejected) is a concrete systems contribution not present in the base paper, and is a more defensible use of blockchain than logging alone.

- **Staking/slashing as an economic ablation:** measure whether adding real (or simulated) economic stakes changes the induced pool of participating clients' behavior in a testbed — this is closer to a mechanism-design/game-theoretic evaluation than a pure systems one, and BlockFLA already provides a template for the empirical methodology.
- **Dispute-window false-positive handling:** quantify how large a challenge window is needed to keep the false-slash rate acceptable given the adapted one-class authentication layer's own false-positive rate (Section 5.3) — a direct, testable link between two components of the system that does not appear to be evaluated together in the reviewed literature.
- **Provenance passport:** lowest-risk, best-precedented blockchain use case here; likely worth implementing as working infrastructure and describing, without needing deep empirical evaluation to justify its inclusion.

7. Explicit Assumptions

- Simulated clients represent distinct organizations (fabs / inspection stations) with genuinely non-IID local defect-category distributions, following the pattern of FedEDAuth's own simulated repair-shop framing.
 - A small number of simulated clients (on the order of 5–15) is used for the primary experiments, reflecting both the realistic number of semiconductor supply-chain partners in a consortium and practical training-time constraints, rather than FedEDAuth's 50-client simulation.
 - The authentication server's golden reference dataset is assumed to be curated by a neutral, trusted third party (mirroring FedEDAuth's own assumption), and its potential compromise is treated as future work, not evaluated directly.
 - On-chain components are prototyped on a public testnet for development speed; production-grade permissioned deployment is discussed but not required to be built.
 - The dataset used for the anomaly-detection task is either the IDA/ICME 2026 dataset (pending confirmed public access) or MIIC with an explicitly stated small-sample-size limitation.
 - “Rollback” refers to reverting the deployed global model to a previous checkpoint's weights; it does not imply modifying already-distributed copies of a poisoned model outside the system's own control.
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8. Evaluation Plan

8.1 Metrics

Component	Metrics
Anomaly detection model	Precision, recall, F1, AUC-ROC on held-out defect imagery; per-defect-category breakdown.
Authentication layer	False positive / false negative rate on suspicion classification; separation margin between clean and poisoned client scores (as in the base paper's Figure 5-style plot).
Rollback mechanism	Detection latency (rounds until trigger), post-rollback accuracy recovery vs. clean baseline, storage/communication overhead vs. naive full-checkpointing and vs. Crab-style selective history.
Culprit attribution	Precision/recall of correctly identifying the actual poisoned client(s) within a triggered window; compute cost vs. full Shapley baseline.
Blockchain layer	Transaction latency/throughput, on-chain storage cost (bytes/gas per round), dispute-window false-slash rate.

8.2 Baselines to compare against

- Centralized (non-federated) training — upper bound.
- Local-only training per client — lower bound.
- Plain FedAvg, no authentication, no rollback.
- FedAvg + per-round-only FedEDAuth-style authentication (no drift tracking) — isolates the value added by multi-round drift detection specifically.

- FedAvg + per-round authentication + full-history recovery (FedRecover-style) — isolates the value added by windowed/selective checkpointing versus full-history methods.
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9. Honest Assessment: Novelty, Feasibility, and Scope Risk

This section restates, in report form, the candid evaluation given during the design discussion, because it materially affects what should and should not be attempted.

9.1 Novelty risk

Nearly every individual mechanism proposed here has close, recent, published prior art: FedEDAuth for authentication, Crab/FedRecover/DP-RTFL for checkpoint-based recovery, Leash-FL specifically for blockchain-anchored FL rollback, and BlockFLA specifically for blockchain-based punishment of FL attackers. None of these individually constitute a defensible new contribution on their own. The one combination that does not appear to already exist in the reviewed literature is: (a) using the authentication layer's own suspicion statistics, tracked across a rolling window rather than per-round, as the rollback trigger itself, and (b) reusing that same triggered window to make culprit attribution cheap. This narrow combination, not the individual pieces, is the project's actual claim to novelty and should be stated as such explicitly in any paper draft.

9.2 Scope risk

The full system as discussed touches seven substantial sub-problems: anomaly-detection modeling, FL orchestration, one-class authentication adaptation, drift-triggered rollback, windowed culprit attribution, multi-validator decentralized authentication, and staking/slashing/dispute mechanics. Attempting to build and rigorously evaluate all seven to the standard of the papers cited in Section 3 is not realistic within a single final-year project timeline. Spreading effort evenly across all of them risks a shallow, unconvincing implementation of each — the specific failure mode a reviewer at IEEE Access level would flag as “a systems demo with too many undercooked parts.”

9.3 Recommended scope

Build and rigorously evaluate **Section 5.4/5.5 (drift-triggered rollback and windowed attribution)** as the core, benchmarked contribution, with full baseline comparisons per Section 8.2. Implement the remaining layers — the vision model, FedAvg orchestration, one-class authentication adaptation, and a basic blockchain layer (Section 5.6, prototyped on a public testnet) — as working supporting infrastructure, described in the report and demonstrated end-to-end, but not each individually benchmarked against its own literature to the same depth. This keeps the system genuinely end-to-end and demonstrable (satisfying the “working product” requirement) while keeping the empirical/research burden concentrated on one defensible, evaluable claim (satisfying the “research paper level understanding” requirement).

10. Summary of Open Decisions

Decision	Options	Status
Dataset	MIIC (confirmed access, few anomalies) vs. IDA/ICME 2026 (richer, access unconfirmed) vs. hybrid/augmented MIIC	Open — verify IDA/ICME 2026 access first
FL aggregation scheme	FedAvg vs. Cyclic Weight Transfer (sequential) vs. hybrid	Decided: FedAvg, pending contrary evidence
Core evaluated contribution	Drift-triggered rollback vs. equal depth across all components	Decided: drift-triggered rollback + windowed attribution
Blockchain platform	Public/Ethereum-style testnet vs. permissioned/Hyperledger-style consortium	Open — recommendation: prototype on public testnet, discuss permissioned as production path
Authentication adaptation depth	Full evaluated sub-contribution vs. working-but-not-benchmarked	Open — depends on remaining timeline after core contribution is built
Client count for simulation	5–15 (repair-shop-style) vs. 50 (matching base paper)	Recommended: 5–15, for tractability and clearer per-client attribution signal

11. References

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